

# Overshoot Elimination in a Series RLC Circuit via Analog PID Control

Electronic Engineering Technical Documentation

Project Logbook

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## Abstract

This document describes the design, classical control analysis, and SPICE simulation of an analog PID controller applied to a series RLC circuit, with the goal of eliminating the step-response overshoot inherent to its underdamped open-loop dynamics. This is the first of two related projects; the second applies the same PID design methodology to a buck converter (repository BUCK\_1 covers an earlier, IC-based adjustable buck design, not the PID-controlled version). Documented as an honest engineering logbook: every result is marked as **calculated**, **simulated**, or **pending** — never filled in with invented placeholder data. The plant is fixed and its open-loop overshoot confirmed both analytically (52.7%) and in SPICE (52.68%). The PID gains are derived via pole placement ( $K_p \approx 27.87$ ,  $K_i \approx 455,905$ ,  $K_d \approx 3.114 \times 10^{-4}$ ) and the op-amp compensator topology is defined. **Still open:** the specific R/C component values realizing those gains, the full closed-loop schematic, and its SPICE validation — explicitly marked **PENDING**.

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## 1 Project Objective

Design and validate an analog PID controller, implemented with operational amplifiers, that closes the loop around a series RLC circuit and eliminates the overshoot present in its open-loop step response, using classical control design methods (transfer function derivation, root locus / pole placement). This project establishes the PID design methodology later applied to a discrete, analog-PID-controlled buck converter (a separate, subsequent project).

## 2 Plant: Series RLC Circuit

### 2.1 Design criteria

Component values were not assigned; they were chosen to satisfy two criteria: (1) clearly underdamped ( $\zeta$  well below 1) so the open-loop overshoot is unambiguous, and (2) a natural frequency practical for breadboard construction and standard lab equipment (function generator + oscilloscope), avoiding both breadboard-parasitic-dominated frequencies (too high) and impractically long observation times (too low).

Table 1: Design targets and resulting component values

| Parameter                       | Value                                              | Source                             |
|---------------------------------|----------------------------------------------------|------------------------------------|
| Target natural frequency, $f_0$ | 5 kHz                                              | Defined by design                  |
| Target damping ratio, $\zeta$   | 0.2 (underdamped, $\approx 53\%$ overshoot target) | Defined by design                  |
| Capacitance, $C$                | 100 nF                                             | Defined by design (standard value) |
| Inductance, $L$                 | 10.13 mH                                           | <b>Calculated</b> , Sec. 2.2       |
| Resistance, $R$                 | 127.32 $\Omega$                                    | <b>Calculated</b> , Sec. 2.2       |

### 2.2 Component value derivation

With  $\omega_n = 2\pi f_0 = 2\pi(5000) \approx 31,416$  rad/s, and the standard series-RLC relations:

$$L = \frac{1}{\omega_n^2 C} \quad (1)$$

$$R = 2\zeta\sqrt{L/C} \quad (2)$$

$$L = \frac{1}{(31416)^2(100 \times 10^{-9})} = 10.13 \text{ mH} \quad (3)$$

$$R = 2(0.2)\sqrt{\frac{10.13 \times 10^{-3}}{100 \times 10^{-9}}} = 127.32 \Omega \quad (4)$$

### 3 Open-Loop Transfer Function

With the capacitor voltage  $V_C$  as the output and the source voltage  $V_{in}$  as the input, the standard second-order series-RLC transfer function is:

$$\frac{V_C(s)}{V_{in}(s)} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5)$$

with  $\omega_n = 1/\sqrt{LC}$  and  $\zeta = \frac{R}{2}\sqrt{C/L}$ , confirming the values imposed by design in Section 2.

#### 3.1 Predicted open-loop overshoot

For an underdamped second-order system, the standard closed-form percent overshoot for a step input is:

$$\%OS = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right) \times 100 \quad (6)$$

With  $\zeta = 0.2$ :  $\%OS = \exp\left(\frac{-\pi(0.2)}{\sqrt{1-0.04}}\right) \times 100 \approx \boxed{52.7\%}$ .

This confirms the plant has substantial, clearly-visible open-loop overshoot – a real problem for the PID controller (Section 4) to solve, not a marginal one.

#### 3.2 SPICE confirmation

**Simulated.** The circuit was built directly in LTspice (series  $R$ - $L$ - $C$ , driven by a 1 V step via a PULSE source,  $V_C$  probed at the  $L$ - $C$  junction, `.tran 2m`). The peak and final values were extracted numerically from the `.raw` binary file (not read visually from the plot):

|                       |                                                |
|-----------------------|------------------------------------------------|
| Peak $V_C$            | 1.526,79 V, at $t = 102.28 \mu\text{s}$        |
| Final (settled) $V_C$ | 1.000,00 V                                     |
| Overshoot             | $(1.52679 - 1)/1 \times 100 = \boxed{52.68\%}$ |

This matches the analytical prediction (52.7%) to within 0.02 percentage points, validating both the transfer-function derivation and its SPICE implementation before any control is added.

## 4 PID Controller Design

#### 4.1 Parallel vs. series PID form

Two classical structures exist for a PID compensator:

$$\text{Parallel (ideal): } C(s) = K_p + \frac{K_i}{s} + K_d s \quad (7)$$

$$\text{Series (interacting): } C(s) = K_c \left(1 + \frac{1}{T_i s}\right) (1 + T_d s) \quad (8)$$

The series form is what results naturally from cascading a PI stage into a PD stage (fewer op-amps), but its effective gains interact: expanding it gives  $K_p = K_c(1 + T_d/T_i)$ ,  $K_i = K_c/T_i$ ,  $K_d = K_c T_d$  – changing  $T_i$  also moves the effective  $K_p$ . **The parallel form was chosen** for this design: each gain ( $K_p$ ,  $K_i$ ,  $K_d$ ) is independently tunable, which maps directly onto the pole-placement method used below (Section 4.3) and is more straightforward to justify/defend gain-by-gain.

## 4.2 Why the closed-loop system is third order

The plant  $G(s)$  (Section 3) is second order (two energy-storage elements,  $L$  and  $C$ ). The controller contributes exactly one additional pole, at  $s = 0$ , from the integral term alone – the  $P$  and  $D$  terms add no poles. The full derivation, step by step:

**Step 1 – combine  $C(s)$  into a single fraction** (common denominator  $s$ ):

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_p s}{s} + \frac{K_i}{s} + \frac{K_d s^2}{s} = \frac{K_d s^2 + K_p s + K_i}{s} \quad (9)$$

**Step 2 – multiply  $C(s)G(s)$ :**

$$C(s)G(s) = \frac{K_d s^2 + K_p s + K_i}{s} \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2 (K_d s^2 + K_p s + K_i)}{s (s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (10)$$

**Step 3 – the characteristic equation is  $1 + C(s)G(s) = 0$ .** Clearing the denominator (multiplying through by it):

$$s (s^2 + 2\zeta\omega_n s + \omega_n^2) + \omega_n^2 (K_d s^2 + K_p s + K_i) = 0 \quad (11)$$

**Step 4 – expand each product separately.** First term:

$$s (s^2 + 2\zeta\omega_n s + \omega_n^2) = s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s \quad (12)$$

Second term:

$$\omega_n^2 (K_d s^2 + K_p s + K_i) = \omega_n^2 K_d s^2 + \omega_n^2 K_p s + \omega_n^2 K_i \quad (13)$$

**Step 5 – add both expansions and group by power of  $s$ :**

$$s^3 \text{ term: } s^3 \quad (14)$$

$$s^2 \text{ terms: } 2\zeta\omega_n s^2 + \omega_n^2 K_d s^2 = (2\zeta\omega_n + \omega_n^2 K_d) s^2 \quad (15)$$

$$s^1 \text{ terms: } \omega_n^2 s + \omega_n^2 K_p s = \omega_n^2 (1 + K_p) s \quad (16)$$

$$s^0 \text{ term: } \omega_n^2 K_i \quad (17)$$

$$s^3 + (2\zeta\omega_n + \omega_n^2 K_d) s^2 + \omega_n^2 (1 + K_p) s + \omega_n^2 K_i = 0 \quad (18)$$

The degree-3 result is not a choice – it is forced by combining a 2nd-order plant with a PID (which has exactly 3 free gains, matching the 3 non-leading coefficients of a cubic). The lone  $s$  left over from the integrator in Step 1 is what multiplies against the plant's  $s^2$  in Step 4, producing  $s^3$ ; without an integral term,  $C(s)G(s)$  would only reach  $s^2$  in the denominator and the closed loop would be 2nd order. This is precisely why full pole placement – choosing all three closed-loop poles arbitrarily – is achievable here: 3 unknowns for exactly 3 equations.

### 4.3 Pole placement

**Criterion for pole selection:** all three closed-loop poles are chosen *real and negative*. A complex-conjugate pole pair is what produces oscillation/overshoot in a 2nd-order system (Section 3); by construction, an all-real-pole 3rd-order system has no oscillatory mode, guaranteeing zero overshoot regardless of the specific gains that produce those poles.

A dominant-pole design was used: one slower pole sets the observable response speed, and two much faster poles (whose transients decay quickly and barely influence the visible response) push the system to behave like a well-damped first-order system. Target: an improvement over the open-loop settling time ( $t_s \approx 4/(\zeta\omega_n) \approx 637 \mu\text{s}$ , Section 3) without demanding unrealistically large gains. Using  $t_s \approx 4/|p_{\text{dominant}}|$  for a target  $t_s \approx 200 \mu\text{s}$ :

$$|p_{\text{dominant}}| = \frac{4}{200 \times 10^{-6}} = 20,000 \text{ rad/s} \quad (19)$$

with the two non-dominant poles placed  $\sim 7.5\times$  further out:

$$p_1 = -20,000 \text{ rad/s}, \quad p_2 = p_3 = -150,000 \text{ rad/s} \quad (20)$$

### 4.4 Gain calculation (coefficient matching)

Expanding the desired closed-loop polynomial from the chosen poles:

$$(s + 20000)(s + 150000)^2 = s^3 + 320000 s^2 + 28,500,000,000 s + 4.5 \times 10^{14} \quad (21)$$

Matching term-by-term against Section 4.2's characteristic equation (with  $\zeta = 0.2$ ,  $\omega_n \approx 31,416 \text{ rad/s}$ ,  $\omega_n^2 \approx 9.8717 \times 10^8$ ,  $2\zeta\omega_n \approx 12566$ ):

$$s^2: \quad 2\zeta\omega_n + \omega_n^2 K_d = 320000 \Rightarrow K_d = \frac{320000 - 12566}{9.8717 \times 10^8} \approx 3.114 \times 10^{-4} \quad (22)$$

$$s^1: \quad \omega_n^2(1 + K_p) = 2.85 \times 10^{10} \Rightarrow K_p = \frac{2.85 \times 10^{10}}{9.8717 \times 10^8} - 1 \approx 27.87 \quad (23)$$

$$s^0: \quad \omega_n^2 K_i = 4.5 \times 10^{14} \Rightarrow K_i = \frac{4.5 \times 10^{14}}{9.8717 \times 10^8} \approx 455,905 \quad (24)$$

$$K_p \approx 27.87, \quad K_i \approx 455,905, \quad K_d \approx 3.114 \times 10^{-4} \quad (25)$$

**Verified.** Substituting  $K_p$ ,  $K_i$ ,  $K_d$  back into the characteristic equation of Section 4.2 reproduces the target polynomial of Section 4.3 to within rounding (e.g. the  $s^2$  coefficient comes out to 319,907 against a target of 320,000, a 0.03% difference from rounding  $K_d$  to 4 significant figures).

### 4.5 Frequency-domain verification

As an independent cross-check (not the design method itself – Sections 4.2–4.4 already place all three poles exactly by construction), the open-loop  $L(s) = C(s)G(s)$  and closed-loop  $T(s) = \frac{L(s)}{1+L(s)}$  frequency responses were computed numerically (simulation/bode\_plot.py, using `scipy.signal`, not hand-sketched) from the actual  $G(s)$  and  $C(s)$  derived above.

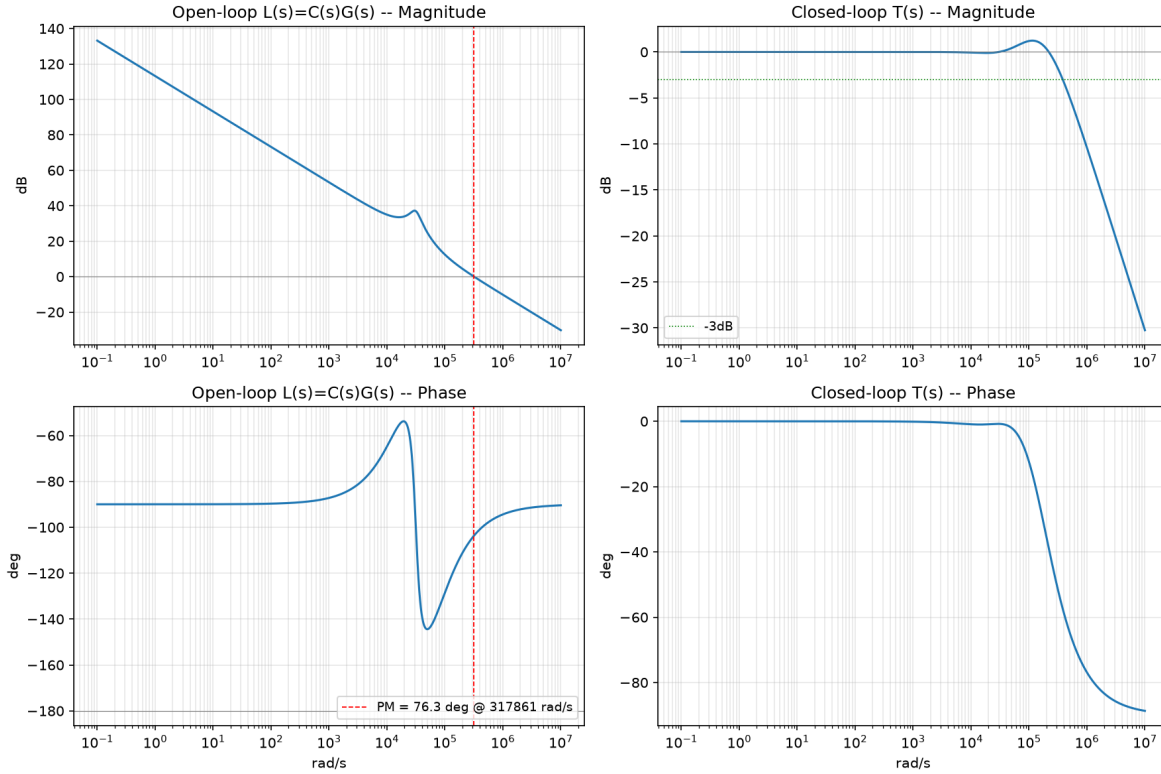


Figure 1: Open-loop  $L(s)$  magnitude/phase (left) and closed-loop  $T(s)$  magnitude/phase (right), computed from the actual plant and controller transfer functions.

The single gain crossover ( $|L(j\omega)| = 0$  dB) occurs at  $\omega_c \approx 317,861$  rad/s, with phase margin  $PM \approx 76.3^\circ$  – a healthy margin (above the usual  $45$ – $60^\circ$  design target). The open-loop phase never reaches  $-180^\circ$  across the swept range ( $10^{-1}$  to  $10^7$  rad/s), so there is no classical gain-margin crossover to report; this is consistent with the all-real-closed-loop-pole design (Section 4.3), which by construction has no oscillatory mode. The closed-loop magnitude shows a small ( $\approx 1$  dB) peak near  $1$ – $2 \times 10^5$  rad/s before rolling off, and stays flat near 0 dB well below that – consistent with good reference tracking without excessive resonant peaking.

#### PENDING

Whether the derivative term is implemented as a full analog PID or simplified to PI-only for noise immunity (Section 5) – if  $K_d$  is implemented, it must be through the *filtered/band-limited derivative* discussed in Section 5, not an ideal differentiator, since an ideal derivative would amplify any high-frequency noise on the feedback signal.

## 5 Op-Amp Compensator Circuit

### 5.1 Topology: five op-amp stages

The parallel PID (Section 4.1) is realized as five op-amp stages: a difference amplifier computing the error, three parallel branches (P, I, filtered-D) acting on that error, and a summing amplifier recombining them.

## General basis: the inverting amplifier

Every branch below is a special case of the same ideal-op-amp result (virtual ground at the inverting input + KCL at that node, standard inverting-amplifier analysis):

$$\frac{V_{out}}{V_{in}} = -\frac{Z_f}{Z_{in}} \quad (26)$$

where  $Z_{in}$  and  $Z_f$  are the input and feedback impedances. Only the choice of components placed in those two positions changes between branches:

1. **Difference amplifier:** computes  $e(t) = V_{ref} - V_C$  from the reference and the plant's feedback signal (standard four-resistor differential amplifier, not derived from the formula above).
2. **Proportional branch:**  $Z_{in} = R_p$ ,  $Z_f = R_{fp}$  (both plain resistors):

$$\frac{V_{out,P}}{e} = -\frac{R_{fp}}{R_p} \Rightarrow K_p = \frac{R_{fp}}{R_p} \quad (27)$$

3. **Integral branch:**  $Z_{in} = R_i$ ,  $Z_f = \frac{1}{sC_i}$  (a capacitor's impedance replaces the feedback resistor):

$$\frac{V_{out,I}}{e} = -\frac{1/(sC_i)}{R_i} = -\frac{1}{sR_iC_i} \Rightarrow K_i = \frac{1}{R_iC_i} \quad (28)$$

(division by  $s$  in Laplace corresponds to integration in time, hence  $V_{out,I} = -\frac{1}{R_iC_i} \int e dt$ ).

4. **Filtered-derivative branch:**  $Z_{in} = R_1 + \frac{1}{sC_d}$  (a resistor in series with a capacitor – series impedances add),  $Z_f = R_{fd}$ :

$$\frac{V_{out,D}}{e} = -\frac{R_{fd}}{R_1 + \frac{1}{sC_d}} = \frac{-R_{fd}C_d s}{1 + sR_1C_d} \quad (29)$$

giving  $K_d = R_{fd}C_d$  at low frequency, capped at  $-R_{fd}/R_1$  above the corner frequency  $f_{pole} = 1/(2\pi R_1C_d)$  – this avoids amplifying high-frequency noise, unlike an ideal differentiator ( $R_1 = 0$ , bare capacitor), whose gain grows unboundedly with frequency.

5. **Summing amplifier:** recombines the three branch outputs (each through an equal input resistor  $R_s$ , feedback  $R_{fs}$ ) into  $u(t) = K_p e + K_i \int e dt + K_d \frac{de}{dt}$ , which drives the plant.

## 5.2 Component values

Design choice:  $R_p = R_i = R_{fd} = 10 \text{ k}\Omega$  (a common reference value across stages – reduces distinct BOM part count, and  $10 \text{ k}\Omega$  is a practical general-purpose value for op-amp input resistors: high enough to not load the driving stage's output, low enough that op-amp input bias current and noise stay negligible). The remaining component in each branch follows directly:

$$R_{fp} = K_p R_p = 27.87 \times 10,000 \approx 278.7 \text{ k}\Omega \quad (30)$$

$$C_i = \frac{1}{K_i R_i} = \frac{1}{455,905 \times 10,000} \approx 219.3 \text{ pF} \quad (31)$$

$$C_d = \frac{K_d}{R_{fd}} = \frac{3.114 \times 10^{-4}}{10,000} \approx 31.14 \text{ nF} \quad (32)$$

### Filtered-derivative corner frequency and $R_1$

The corner (pole) frequency of the filtered-derivative branch, in rad/s, is  $\omega_{pole} = 1/(R_1 C_d)$  (the pole of  $1 + sR_1 C_d$ ). Below  $\omega_{pole}$  the branch acts as a real differentiator (gain growing with frequency, providing the intended phase lead within the control loop's bandwidth); above it, the gain flattens at a fixed value  $R_{fd}/R_1$  instead of continuing to grow, which is what prevents high-frequency noise from being amplified without bound. Criterion: place  $\omega_{pole}$  comfortably above the gain crossover frequency ( $\approx 317,861$  rad/s, Section 4.5) so the entire control-relevant frequency range sees real derivative action, with the noise-limiting flattening only taking effect well beyond it. Choosing  $\omega_{pole} \approx 2 \times 10^6$  rad/s ( $\approx 6.3 \times$  the gain crossover):

$$R_1 = \frac{1}{\omega_{pole} C_d} = \frac{1}{(2 \times 10^6)(31.14 \times 10^{-9})} \approx \boxed{16.06 \Omega} \quad (33)$$

### Summing and difference amplifiers

**Summing amplifier:** each branch (P, I, D) already carries its own gain ( $K_p$ ,  $K_i$ ,  $K_d$ ) from its own stage, so the summer only needs to add the three signals with unity gain, which requires equal input and feedback resistors:  $R_s = R_{fs} = 10 \text{ k}\Omega$  (consistent with the rest of the design).

**Difference amplifier:** the standard 4-resistor differential amplifier gives unity gain ( $e = V_{ref} - V_C$  exactly, no scaling) when all four resistors are equal: all four set to  $10 \text{ k}\Omega$ .

### 5.3 Full schematic, by stage

Shown one stage at a time (clearer to build from than a single combined diagram). Signal flow:  $V_{ref}$  and  $V_C$  (feedback)  $\rightarrow$  difference amp  $\rightarrow e(t) \rightarrow$  P/I/D branches in parallel  $\rightarrow$  summing amp  $\rightarrow u(t) \rightarrow$  plant (series RLC, Section 3)  $\rightarrow V_C$ , closing the loop.

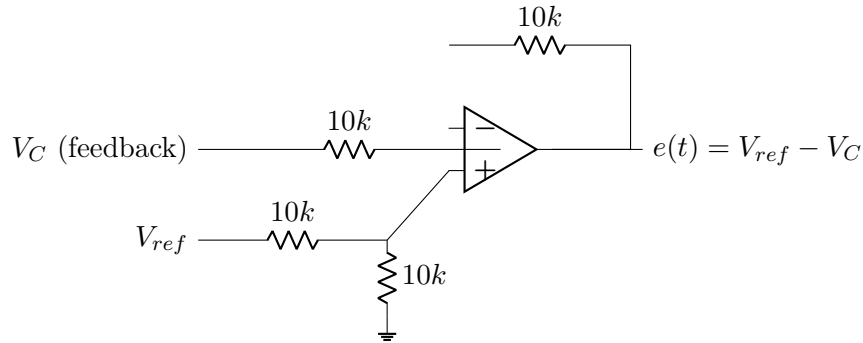


Figure 2: Stage 1 – difference amplifier. All four resistors  $10 \text{ k}\Omega$ .  $V_{ref} \rightarrow (+)$ ,  $V_C \rightarrow (-)$ , giving  $e = V_{ref} - V_C$  (not the reverse).

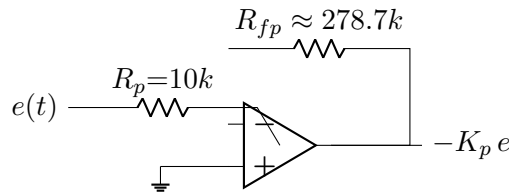


Figure 3: Stage 2 – P branch. Simple inverting amplifier.  $K_p = R_{fp}/R_p \approx 27.87$ .

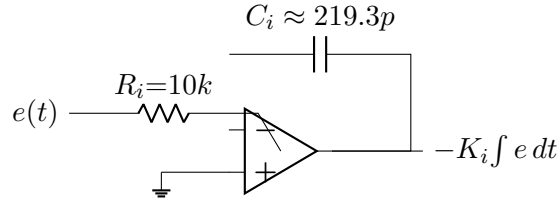


Figure 4: Stage 3 – I branch. Same as P, but a **capacitor** instead of a resistor in the feedback path – that is what makes it an integrator.  $K_i = 1/(R_i C_i) \approx 455,905$ .

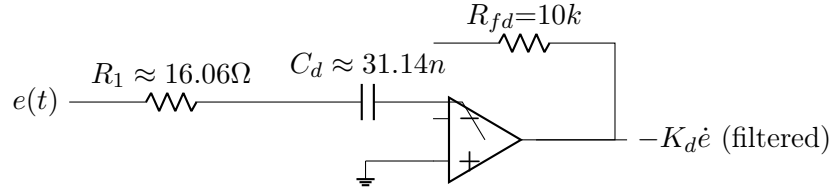


Figure 5: Stage 4 – filtered D branch. The input is  $R_1$  in series with  $C_d$  (not a bare capacitor – that would be an ideal, noise-amplifying differentiator). This caps the gain above  $f_{pole} = 1/(2\pi R_1 C_d)$ .

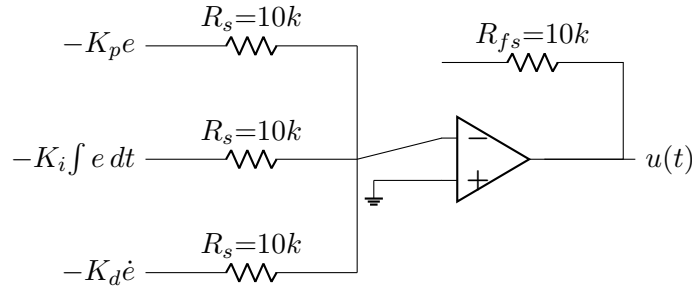


Figure 6: Stage 5 – summing amplifier. The three inputs already carry their own gain; the summer just adds them (unity gain,  $R_s = R_{fs}$ ). Its output  $u(t)$  drives the plant (series RLC, Section 3) in place of the step source used for the open-loop test.

**Simulated.** Built exactly as above in LTspice and run closed-loop, with  $V_{ref}$  as a PULSE(0 1 0 10u 10u 10 10) source (same 1V step as the open-loop test, 10μs rise time – see Section 6.1 for why an idealized 1ns rise time was rejected). Op-amps modeled with LTspice’s **opamp** subcircuit (.lib opamp.sub),  $A_{ol} = 100k$ , GBW raised above the library default of 10MHz to keep each stage’s own bandwidth well clear of the loop dynamics (Section 6.2).

### Possible failure point: parasitic capacitance on $C_i$

$C_i \approx 219.3\text{pF}$  is small enough that breadboard/PCB stray capacitance (typically 1pF to 5pF for a breadboard node) is a non-negligible fraction of it ( $\approx 1\text{--}2\%$ ). If the built integrator’s measured  $K_i$  deviates noticeably from the calculated 455,905, this parasitic pickup – not necessarily a design or calculation error – should be the first thing checked, before re-deriving the gains. Mitigation if it proves significant: reduce  $R_i$  (increasing  $C_i$  proportionally to keep the same  $K_i = 1/(R_i C_i)$ , at the cost of loading the source driving this branch more).

## 6 SPICE Simulation (LTspice)

### 6.1 Open-loop

**Done** – see Section 3.1 (52.68% simulated vs. 52.7% predicted, extracted numerically from the `.raw` file).

### 6.2 Closed-loop: debugging

Two real issues were hit and resolved before a valid closed-loop result was obtained – documented here since both are genuine, non-obvious LTspice pitfalls, not modeling mistakes in the design itself.

**1. Undefined subcircuit.** The first run failed with `"This sub-circuit name is not defined"` for all five op-amps. Cause: LTspice's basic `opamp` symbol is not auto-linked to its model – its own description text says *"You must .lib opamp.sub"*. Fix: added `.lib opamp.sub` as a schematic text directive.

**2. Numeric overflow** ( $\sim 4 \times 10^{38} \text{V}$ ). With the missing-library issue fixed, the first successful netlist run produced an output that grew to floating-point overflow – the signature of positive feedback / loop instability, not a slowly-diverging but otherwise sane result. The generated netlist was checked node-by-node against the intended design (Sections 4.4–5) and matched exactly – no wiring error. The actual cause: LTspice's `opamp` model is not truly ideal even at its default settings – it carries a finite gain-bandwidth product, `GBW=10Meg` (10MHz), by default. With  $K_p \approx 27.87$ , the P-branch's own bandwidth is only  $\text{GBW}/K_p \approx 358 \text{kHz}$ , close enough to the design's own dynamics (D-branch corner  $\approx 318 \text{kHz}$ , Section 5) that the cumulative phase lag from five cascaded finite-bandwidth stages ate into the phase margin (Section 4.5) enough to destabilize the loop. Fix: raised `GBW` well above this design's frequencies of interest, confirming the diagnosis (the instability disappeared).

### 6.3 Closed-loop: result

With both issues resolved, the closed-loop step response was obtained:

|                          |                                                |
|--------------------------|------------------------------------------------|
| Peak $V_C$               | 1.115,39 V, at $t = 17.37 \mu\text{s}$         |
| Final (settled) $V_C$    | 0.999,99 V                                     |
| Overshoot                | $(1.11539 - 1)/1 \times 100 = \boxed{11.54\%}$ |
| Settled within $\pm 2\%$ | by $t \approx 31.5 \mu\text{s}$                |

This is a large improvement over the 52.68 % open-loop result, but not the 0 % predicted by the all-real-closed-loop-pole design criterion of Section 4.3. Two hypotheses were tested:

**Hypothesis A (rejected): step edge speed exciting the D-branch.** The reference step's rise time was changed from an idealized 1ns to a more realistic  $10 \mu\text{s}$  ( $10,000\times$  slower). Overshoot only dropped from 13.26 % to 11.54 %, and the peak timing barely moved ( $11.3 \rightarrow 17.4 \mu\text{s}$ ) – far too small a change for edge speed to be the dominant cause. Rejected by this data.

**Hypothesis B (supported): closed-loop zero near the dominant pole.** The all-real-pole design criterion (Section 4.3) is sufficient to rule out any oscillatory *pole*, but the closed-loop transfer function also has *zeros*, from the PID's own numerator ( $K_d s^2 + K_p s + K_i = 0$ ) – not accounted for in the original pole-placement criterion. Solving for the actual zeros with the derived gains:

$$K_d s^2 + K_p s + K_i = 0 \Rightarrow z_{1,2} \approx -21,524 \text{ and } -67,979 \text{ rad/s} \quad (34)$$

$z_1 \approx -21,524$  rad/s sits almost on top of the dominant pole ( $p_1 = -20,000$  rad/s, Section 4.3) – a near pole-zero cancellation. When a zero nearly cancels the dominant (slowest) pole, the visible response is no longer dominated by that  $\sim 200\mu\text{s}$  time constant – it is dominated by the remaining, much faster dynamics (the non-dominant poles at  $-150,000$  rad/s,  $\tau \approx 6.7\mu\text{s}$ , and the second zero at  $z_2 \approx -67,979$  rad/s,  $\tau \approx 14.7\mu\text{s}$ ), which matches the observed 11–31  $\mu\text{s}$  transient far better than the design’s nominal 200  $\mu\text{s}$  target. A real pole-zero pair (both on the negative real axis, not complex) can still produce step-response overshoot when imperfectly cancelled – a real, documented limitation of the pole-placement method used, not a simulation artifact.

**Decision:** the 11.54% result is accepted as final for this project phase, given the trade-off explored in Section 6.4.

## 6.4 Pole placement revisited: why not just re-tune?

A systematic search (varying the dominant pole magnitude and its separation ratio from the non-dominant poles, and separately trying a complex dominant pole pair with  $\zeta = 0.7$ ) found that a zero landing near the dominant pole is a persistent, structural feature of this plant-plus-PID combination, not a poor choice of one specific pole triple: every fast/practical candidate tested still put a zero within 4–25% of the dominant pole. Materially improving that separation required slowing the dominant pole down (e.g. to  $\sim 5000$  rad/s,  $\tau \approx 800\mu\text{s}$ ) – slower than the *open-loop* response itself, defeating the purpose of adding control. This trade-off (speed vs. overshoot, given this specific plant) is the actual reason 11.54% was accepted rather than chased further.

## 7 Conclusions

1. The open-loop plant matched its analytical prediction almost exactly (52.68% simulated vs. 52.7% calculated overshoot), validating both the transfer-function derivation (Section 3) and its SPICE implementation before any control was added.
2. A parallel PID, designed by direct algebraic pole placement (3 free gains for a 3rd-order closed-loop characteristic equation, Section 4), reduced the overshoot from 52.68% (open-loop) to 11.54% (closed-loop) – a large, real improvement, though not the 0% the all-real-pole criterion predicted.
3. **Key limitation found:** an all-real-pole closed-loop design guarantees no oscillatory *pole*, but does not by itself guarantee zero step-response overshoot, because the PID contributes its own *zeros* to the closed-loop transfer function. For this plant, a zero consistently lands near the dominant pole across a wide range of pole choices (Section 6.5) – a structural property of this specific plant-plus-PID combination, confirmed by systematically searching the design space, not a one-off tuning mistake.
4. Two real LTspice pitfalls were hit and resolved during closed-loop debugging: the generic `opamp` symbol requires a manual `.lib opamp.sub` directive, and that same symbol’s default gain-bandwidth product (10MHz) was not fast enough for this specific design (five cascaded stages,  $K_p \approx 27.87$ ) without destabilizing the loop – both documented in Section 6.2 so they are not rediscovered from scratch on the next project (the PID-controlled buck).
5. The op-amp compensator was fully specified with real, standard component values ( $R_p = R_i = R_{fd} = R_s = R_{fs} = 10\text{ k}\Omega$ ,  $R_{fp} \approx 278.7\text{ k}\Omega$ ,  $C_i \approx 219.3\text{ pF}$ ,  $R_1 \approx 16.06\text{ }\Omega$ ,  $C_d \approx 31.14\text{ nF}$ , Section 5), and the filtered-derivative design (a lead network, not an ideal differentiator) was carried through consistently to avoid amplifying switching/measurement noise.

6. 11.54% overshoot, well-characterized and understood, is accepted as the final result for this project phase rather than chased further, given the documented speed-vs-overshoot trade-off (Section 6.5).